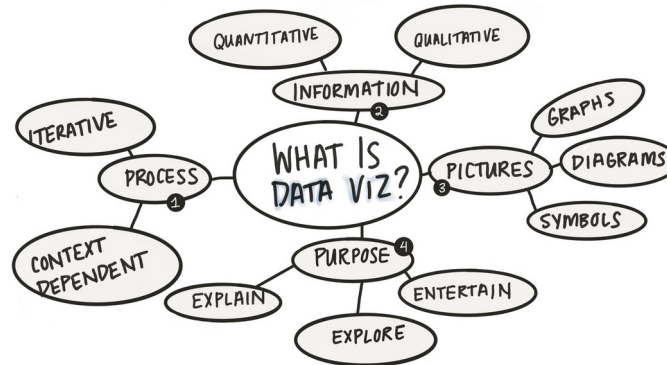


Lecture 15

Visualization Intro



Next Three Weeks

- ❖ Data visualization
 - Types of plots, plot features (colors, shapes, etc.), modes of display
 - Building plots using Pandas and Seaborn
- ❖ Modes of communication
 - Slides, posters, reports, blog posts, academic papers
- ❖ Effective Communication
 - Strategies for designing and presenting information

Today

Data visualization

- ❖ Why care?
- ❖ Good plot – bad plot

Project Details

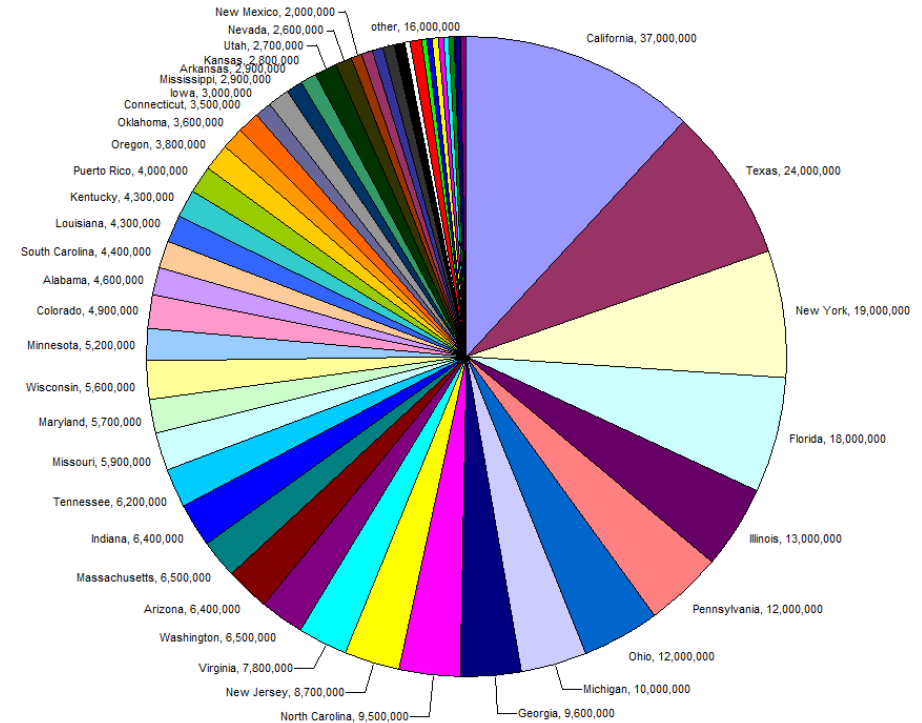
- ❖ Work in randomized teams of four
- ❖ Each team will choose a dataset and a scenario (first come, first served)
- ❖ Deliverables
 - Data exploration and problem description
 - Investigating questions
 - Forming new hypotheses
 - Final report + presentation
- ❖ Teams and topics released Wednesday

Why care?

- ❖ There are many ways to visualize the same data
- ❖ But not all convey the same information
- ❖ Visualizations can be key to telling a compelling story
- ❖ ...or confusing/frustrating the reader

Why care?

- ❖ **Good visualizations** clearly and accurately convey the key information in the data
- ❖ **Bad visualizations** obfuscate the data, intentionally or otherwise

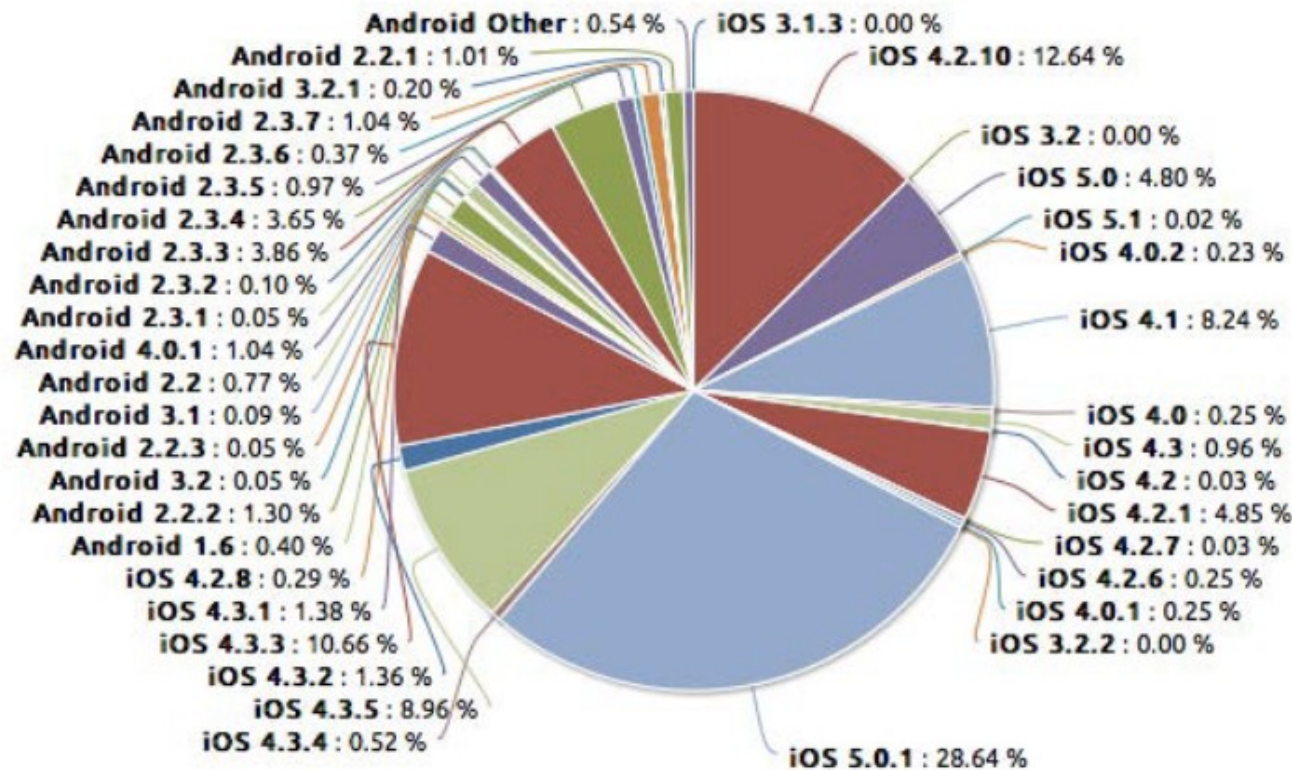


Data Viz Use Cases

- ❖ Exploratory
 - An analyst may use visualizations to investigate a dataset
- ❖ Descriptive
 - Convey information to an audience
- ❖ Rhetorical
 - Use a visualization to try to convince an audience

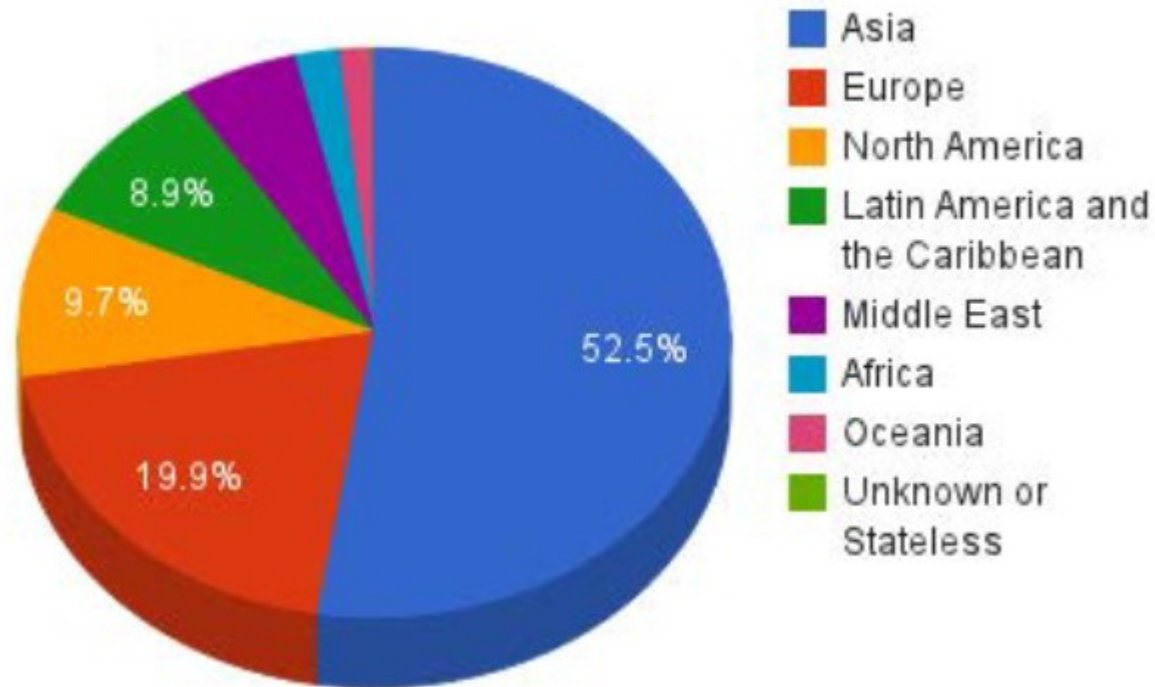
Good or bad?

Crashes by OS Version Normalized (12/1 - 12/15)

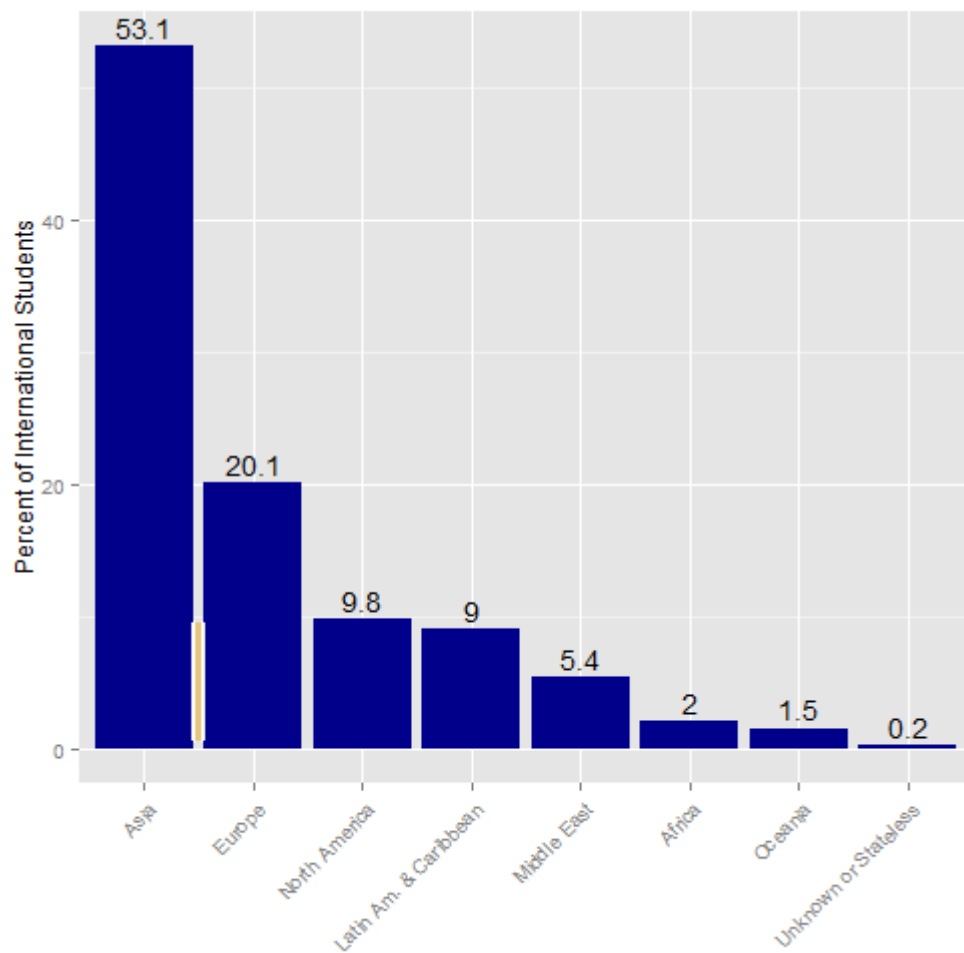


Good or bad?

MIT International Graduate Students



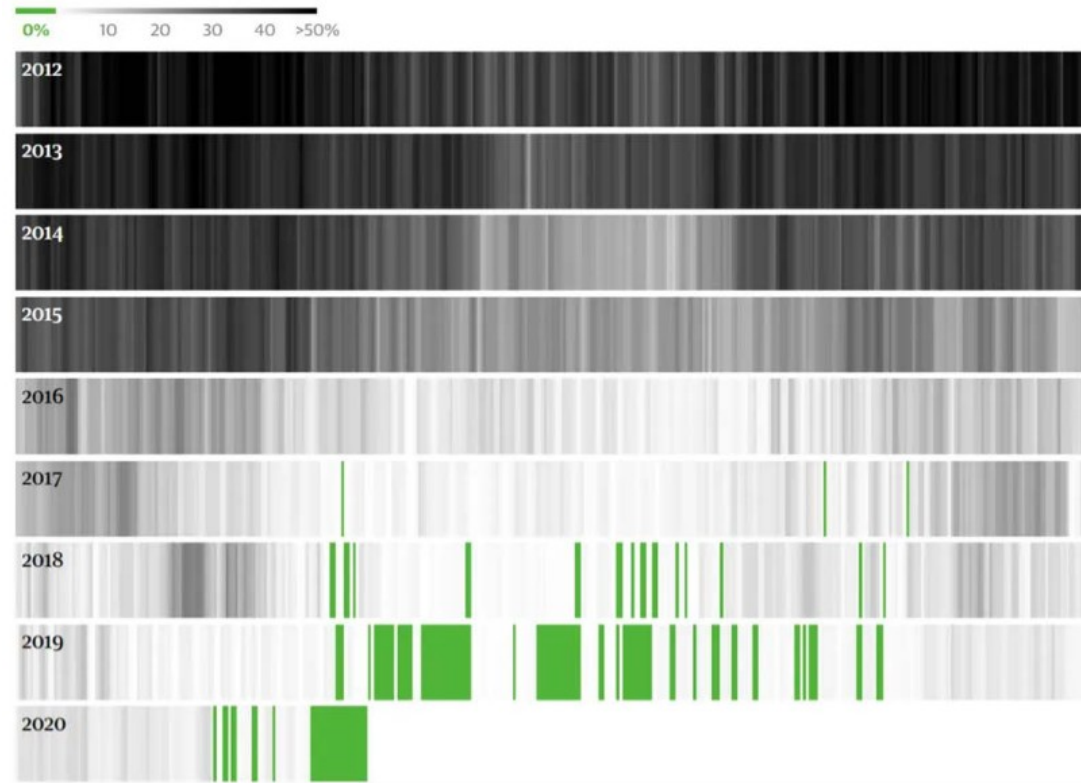
Good or bad?



Good or bad?

Britain is rapidly phasing out coal

Daily share of Britain's power generated by burning coal



Source: Drax Electric Insights. Last updated at 6:45pm on 28 April

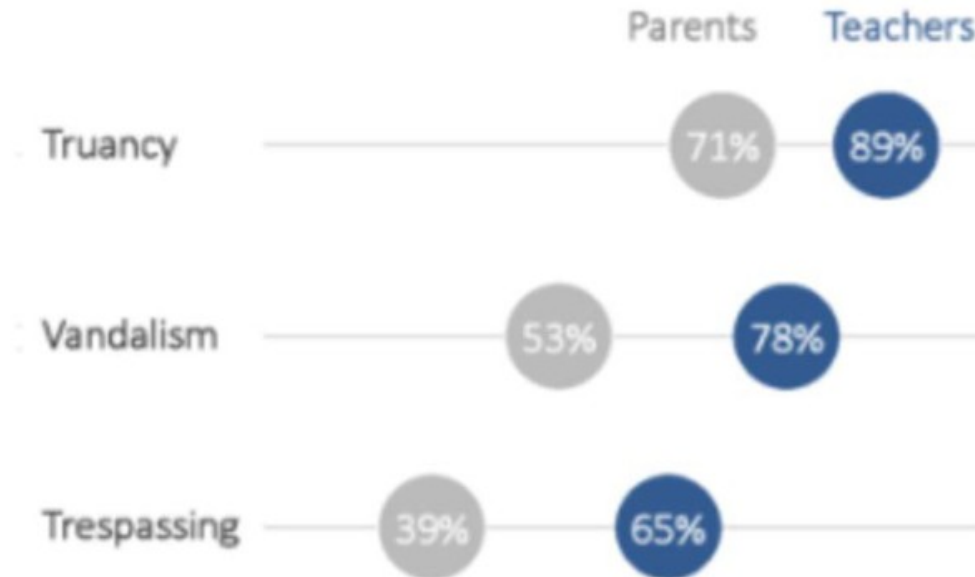
Good or bad?



Good or bad?

2014-15 school safety survey results

Teachers were more likely than parents to indicate that truancy, vandalism, and trespassing are "pressing needs" that should be addressed.



Next Time

- ❖ Plotting in Seaborn